Team 2 - Billy’s Amazing Team

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Introduction (5a):

As we are conducting a Task-Based User Evaluation, we will follow the 7 steps needed to carry out this process. The seven steps are listed below:

Recruit Participants:
The first step is to recruit participants. Ideally these participants will be as representative of the end user as possible, however this is not always the case and as long as the user will see the game in the same way as the end users, they will be able to provide an adequate evaluation for our project. We recruited our participants from our ENG1 Cohort, as they were easily available for the evaluation and represented the end user well enough. We recruited 5 participants, as according to many researchers, this figure is appropriate for our small project size. All participants gave informed consent via reading and signing the informed consent forms we provided to them before they took part.

Defining the Tasks:
We decided to use our tasks to guide the user through our game, their tasks would be to first construct one building, then to try and increase the satisfaction bar however they see fit and finally to react to an event how they see fit, which would guide them to the end of the game.

Preparing the Environment:
We conducted the evaluation within the floor 2 software laboratory of the Ian Wand building. This was due to it being convenient, as all the participants were already in this room during the time slot chosen for the evaluations. Additionally, it allowed us to control the environment between participants, offering them the same device to use and seating arrangement. On top of this, the environment chosen is a good representation of an example location that the end user might use our product in, which adds additional value to the evaluation.

Planning How the Tasks Will be Run:
All the participants will be present at the evaluation, as they have the same ENG1 timetable as us due to being in the same cohort. We have also tested the tasks on a group member to ensure that it will run correctly at the evaluation, avoiding wasted time fixing issues.

Carrying Out the Tasks:
We used a think aloud protocol (CVP) with an observer writing down the thoughts stated by our user, this meant one evaluator would run the tasks and the scribe would purely write down the results, minimising the risk of missing key information.

Compiling the Data:
The compilation of usability problems identified by the users can be seen in the following section of this document. We have created a table that lists the usability problems as well as the severity ratings of these problems as rated by the participants themselves.

Making Design Recommendations:
After the evaluations and compiling the data, we will use our findings to improve upon the design of our system via a redesign, which aims to avoid the problems identified by users. The most severely rated, and most common problems identified will be the priority in terms of recommendations made to the implementation team.

Problems found (5b):

Below is a table of the usability problems made apparent by the users during testing. The severity uses the ratings from the Think Aloud Protocols (1 cosmetic, 2 minor, 3 major, 4 catastrophic).

User (C1-C5)	Problem	Severity
All users	Cannot differentiate/remember buildings	3
C1	Does not understand differences between student thoughts	2
C1	Does not know what ‘0’ on top of building means CONT	2
C1/C2/C3	Satisfaction bar is not labelled as such (function not obvious)	4
C1	Not enough slots for buildings	2
C1	Cannot make game fullscreen	1
C1	Construction not progressing while paused is not obvious	2
C1/C2	Satisfaction bar arrow meaning not obvious	3
C2/C5	Cannot distinguish items on tool button	3
C2	Does not know how location slot will affect building’s utility	2
C2	Did not notice events happening	2
C2/C4	Needs to make connection between actions and goal, unsure how actions affect satisfaction	2
C2	End screen not helpful for feedback on performance	1
C3/C4	Too much text in thoughts box	1
C4	Unsure how to pause game (assumes space bar)	2
C4	Game too overwhelming to learn all the UI and play at the same time at the beginning, needs a grace period for learning	3
C4	Cannot understand how to demolish a building	2
C5	No warning before demolition	2
C5	Icons are too small	1
C5	Leaderboard text is formatted awkwardly	1